

WifeSignal

Arie Meir designed and built a three-colour signalling device that lets one person in the house tell another how urgently they are needed, without a phone notification. A custom iPhone app, a Python bridge, BLE firmware, a fabricated PCB and a generated enclosure, joined by a one-byte protocol.

The problem

The office is a separate building. Work happens there with music on and the windows closed, and the phone face-down and unchecked for hours — which is the point, and which makes it a sealed box. So **call your mother back this week** and **the delivery driver is leaving in thirty seconds** arrive through the same silent channel and get read at the same time: eventually. Texting and hoping fails exactly when it matters; walking over costs an interruption either way.

The approach

The missing piece was never notification — it was **priority, expressed before the interruption happens**. Three buttons in an app map to three lamps on a desk: *Whenever*, *Soon*, *Now*. A Mac already sitting near the lamp acts as the permanent BLE central, because an iPhone cannot reliably hold that role for a device in another building. Pressing the lit button acknowledges it, and the sender's phone shows **Seen**.

The build

Four stages over about two weeks, each earning the next: one channel on protoboard to prove the BLE service and the button semantics; a fabricated two-layer board with three low-side lamp drivers; assembly onto 30 mm illuminated arcade buttons with the microcontroller on removable headers; and a two-part enclosure whose mounting pillars are cut to match notches in the board's own edge — the electrical and mechanical designs share one set of coordinates, written down once.



In service, showing green. The speaker beside it is the reason the project exists.



Rev A board wired to three illuminated arcade buttons, ESP32-C3 on removable headers.

Impact

In daily use. Urgency is now visible in peripheral vision instead of buried in a phone, and the acknowledgement path closes the loop — the sender can see the message landed rather than guessing whether to walk over. An asymmetric, lossy channel became a shared vocabulary of three words.

1 Send

SwiftUI app

Three priority buttons over HTTPS, reachable only on a private tailnet.



2 Bridge

aiohttp + bleak

One file, one event loop. Holds the BLE connection the phone cannot.



3 Drive

ESP32-C3 / NimBLE

One service, one characteristic, one byte driving three lamp channels.



4 Acknowledge

Physical press

Setting the byte to zero is the acknowledgement, notified back.

Engineering architecture

An illegal state that cannot be expressed

Three channels are addressed by *value*, not by bitmask, so "red and green at once" is unrepresentable on the wire. The product rule is enforced by the encoding — there is no validation code for it anywhere, because there is nothing to validate.

Convergence instead of delivery

The server never asks whether a message arrived; it continuously asserts what the lamps should show and re-asserts on every reconnect. A device that reboots or is reflashed mid-signal recovers with no retry queue and no reconciliation logic.

Telling your own echo from a human being

The characteristic is bidirectional, so a write echoed back, a button press and a drifted device all arrive as the same one-byte notification. They are separated by comparing against the state the server already intends — three branches, no timers, no correlation IDs.

The soldered parts win

The drivers on the board are PNP; the schematic says NPN — an assortment-kit mixup found after assembly. The firmware inverts and the drawing is documented as wrong, rather than reworking a functioning board to match a diagram.

Technical scope

SwiftUI iOS app · Python asyncio HTTP API · BLE GATT protocol design · ESP32-C3 / NimBLE firmware · KiCad schematic + PCB + gerbers · Parametric Fusion 360 enclosure generator · launchd + Tailscale deployment

Design principle: make illegal states unrepresentable at the lowest layer, converge on intent rather than delivering messages, and detect the symptom of every failure that would otherwise be silent.